What is a database?
A. A network cable
B. A programming language
C. A collection of organized data
D. An operating system
ANSWER: C

In MySQL, which command creates a database?
A. MAKE DATABASE
B. NEW DATABASE
C. ADD DATABASE
D. CREATE DATABASE
ANSWER: D

Which command selects a database for use?
A. SELECT DATABASE database_name
B. USE database_name
C. OPEN database_name
D. CONNECT database_name
ANSWER: B

What is a table in a relational database?
A. A query editor
B. A database user
C. A backup file only
D. A collection of rows and columns
ANSWER: D

What does a database schema describe?
A. Only the password of a user
B. The monitor resolution
C. The structure of database objects
D. The network route
ANSWER: C

Which MySQL statement removes a database and its objects?
A. DELETE DATABASE
B. REMOVE DATABASE
C. DROP DATABASE
D. CLEAR DATABASE
ANSWER: C

Which command creates a table named students?
A. NEW TABLE students
B. ADD students
C. MAKE students TABLE
D. CREATE TABLE students (...)
ANSWER: D

Which statement adds a new row to a table?
A. ALTER TABLE
B. UPDATE
C. DROP TABLE
D. INSERT INTO
ANSWER: D

Which SQL command changes an existing table's structure?
A. RENAME COLUMN ONLY
B. CHANGE DATABASE
C. MODIFY DATABASE
D. ALTER TABLE
ANSWER: D

Which SQL command can create a MySQL account?
A. MAKE LOGIN
B. NEW ACCOUNT
C. ADD LOGIN
D. CREATE USER
ANSWER: D

Which SQL statement retrieves data from a table?
A. FETCH TABLE
B. GET DATA
C. SELECT
D. SHOW ROWS
ANSWER: C

In SELECT * FROM students, what does the asterisk mean?
A. Select every column
B. Select only the first row
C. Count the rows
D. Delete all columns
ANSWER: A

Which clause filters rows before they are returned?
A. HAVING
B. GROUP BY
C. ORDER BY
D. WHERE
ANSWER: D

Which operator checks whether a value is within a range?
A. GROUP
B. JOIN
C. BETWEEN
D. ORDER
ANSWER: C

Which operator can match a text pattern with wildcards?
A. PATTERN
B. LIKE
C. SEARCH
D. MATCHES ONLY
ANSWER: B

Which clause sorts query results?
A. ORDER BY
B. SORT
C. SORT BY
D. ARRANGE
ANSWER: A

What does ORDER BY age DESC do?
A. Sorts age from lowest to highest
B. Deletes duplicate ages
C. Sorts age from highest to lowest
D. Groups ages
ANSWER: C

What is the purpose of an INNER JOIN?
A. Creates a new database
B. Returns matching rows from both tables
C. Deletes unmatched rows
D. Returns every possible row regardless of match
ANSWER: B

Which keyword specifies the columns used to match joined rows?
A. ON
B. USING AS
C. MATCH
D. WHERE ONLY
ANSWER: A

Which query is used to return selected columns from employees?
A. GET name, role employees;
B. SHOW employees(name, role);
C. SELECT name, role FROM employees;
D. FETCH name AND role IN employees;

ANSWER: C

Which condition selects students whose marks are at least 60?
A. WHERE marks >= 60
B. ORDER BY marks >= 60
C. HAVING marks = 60
D. WHERE marks <= 60
ANSWER: A

How can a query test for a missing SQL value?
A. EQUALS NULL
B. IS NULL
C. NULL()
D. = NULL
ANSWER: B

Which clause combines conditions where both must be true?
A. OR
B. AND
C. NOT
D. LIKE
ANSWER: B

What is the result of using OR between two WHERE conditions?
A. Both conditions must always be true
B. The result is sorted
C. A row can match either condition
D. All rows are deleted
ANSWER: C

Which clause returns rows in alphabetical order by name?
A. GROUP BY name ASC
B. WHERE name ASC
C. ORDER name ALPHABETICAL
D. ORDER BY name ASC
ANSWER: D

What does LIMIT 5 do in MySQL?
A. Returns at most five rows
B. Sorts five columns
C. Creates five tables
D. Filters rows with value five
ANSWER: A

Which join returns all rows from the left table and matching rows from the right?
A. INNER JOIN
B. LEFT JOIN
C. RIGHT EXCLUDE JOIN
D. CROSS JOIN only
ANSWER: B

If two tables have a common student_id, what is it often used for in a JOIN?
A. Matching related records
B. Sorting the database files
C. Encrypting passwords
D. Dropping duplicate tables
ANSWER: A

Which SQL clause should appear before ORDER BY in a typical SELECT query?
A. WHERE
B. DROP
C. CREATE
D. GRANT
ANSWER: A

Which statement correctly sorts newest records first using created_at?
A. ORDER BY created_at DESC
B. WHERE created_at DESC
C. GROUP created_at NEWEST

D. ORDER BY created_at ASC
ANSWER: A

Which aggregate function counts rows?
A. COUNT()
B. SUM()
C. ROUND()
D. AVG()
ANSWER: A

Which aggregate function adds numeric values?
A. COUNT()
B. TOTAL()
C. SUM()
D. ADDROWS()
ANSWER: C

What does AVG(price) calculate?
A. Sum of prices
B. Number of prices
C. Highest price
D. Average price
ANSWER: D

Which clause groups rows with the same value?
A. JOIN BY
B. WHERE
C. ORDER BY
D. GROUP BY
ANSWER: D

Which clause filters groups after aggregation?
A. ORDER BY
B. WHERE
C. HAVING
D. LIMIT
ANSWER: C

Which query counts students in each course?
A. COUNT students GROUP course;
B. SELECT course COUNT students;
C. SELECT GROUP course FROM students;
D. SELECT course, COUNT(*) FROM students GROUP BY course;
ANSWER: D

Why is GROUP BY commonly used with COUNT()?
A. To delete repeated rows
B. To calculate counts for each category
C. To connect to MySQL
D. To rename a table
ANSWER: B

Which aggregate function ignores NULL values in its expression?
A. ALTER(column)
B. USE(column)
C. SUM(column)
D. DROP(column)
ANSWER: C

Which clause would select groups with COUNT(*) greater than 10?
A. ORDER BY COUNT(*) > 10
B. WHERE COUNT(*) > 10
C. HAVING COUNT(*) > 10
D. GROUP COUNT(*) > 10
ANSWER: C

Which query calculates total sales per region?
A. SELECT region, SUM(sales) FROM orders GROUP BY region;
B. GROUP orders BY sales SUM region;

C. SELECT region FROM orders HAVING sales;
D. SELECT SUM region FROM orders;
ANSWER: A

What is a subquery?
A. A user account
B. A database server
C. A table backup
D. A query nested inside another SQL query
ANSWER: D

Where can a subquery be used in SQL?
A. Only in user passwords
B. Only after DROP
C. Only in database names
D. In places supported by the SQL statement, such as WHERE
ANSWER: D

Which operator is commonly used to test whether a value matches any value returned by a subquery?
A. ORDER
B. IN
C. GROUP
D. ALTER
ANSWER: B

What does EXISTS test in a subquery?
A. Whether the subquery returns at least one row
B. Whether a database is backed up
C. Whether a table has a primary key only
D. Whether a column is sorted
ANSWER: A

A subquery that refers to a column from the outer query is called what?
A. Correlated subquery
B. Detached table
C. Static schema
D. Cross database
ANSWER: A

Which query pattern finds employees earning above the overall average salary?
A. ORDER BY salary AVG
B. WHERE AVG(salary) > employees
C. GROUP BY salary > AVG
D. WHERE salary > (SELECT AVG(salary) FROM employees)
ANSWER: D

What must a scalar subquery return when used where a single value is expected?
A. At most one row and one column
B. A complete database
C. Exactly ten rows
D. Only text values
ANSWER: A

Which keyword can compare a value against all values returned by a subquery?
A. SCHEMA
B. ALL
C. COLUMN
D. TABLE
ANSWER: B

Why might a subquery be useful?
A. It automatically creates a user
B. It always replaces the database
C. It can calculate a value or set used by an outer query
D. It removes the need for SQL syntax
ANSWER: C

Which aggregate could be placed in a subquery to find the maximum mark?
A. TABLE()

B. ORDER()
C. MAX()
D. JOIN()
ANSWER: C

Which command commonly checks the installed Python 3 version?
A. check python install
B. python3 --version
C. version python --delete
D. python3 --remove
ANSWER: B

Which function displays output in a basic Python program?
A. echo()
B. displayline()
C. writeconsole()
D. print()
ANSWER: D

Which function reads text entered by a user?
A. scanline()
B. gettextline()
C. input()
D. readkey()
ANSWER: C

Which keyword begins a function definition in Python?
A. function
B. define
C. func
D. def
ANSWER: D

How does Python normally mark a block of code?
A. Curly braces only
B. Semicolons only
C. Indentation
D. Line numbers
ANSWER: C

Which is a Python numeric data type?
A. paragraph
B. table
C. int
D. chararray only
ANSWER: C

What is the type of the literal 'Hello'?
A. bool
B. str
C. int
D. float
ANSWER: B

Which Python value represents a Boolean false value?
A. 0False
B. False
C. false
D. FALSE
ANSWER: B

What does type(3.14) return?
A. int
B. str
C. float
D. bool
ANSWER: C

Which is a valid Python comment?

A. ** comment **
B. // This is a comment
C. # This is a comment
D. <!-- comment -->
ANSWER: C

Which Python literal creates an empty list?
A. ()
B. []
C. {}
D. <>
ANSWER: B

Which Python literal creates an empty dictionary?
A. []
B. {}
C. ()
D. dict[]
ANSWER: B

How do you access a dictionary value using key 'name' in variable student?
A. student->name
B. student['name']
C. student(name)
D. student.name
ANSWER: B

Which list method adds one item to the end?
A. extendone()
B. pushback()
C. insertendonly()
D. append()
ANSWER: D

Which list method inserts an item at a specified index?
A. indexadd()
B. insert()
C. appendat()
D. place()
ANSWER: B

How can you get the number of items in a list?
A. len(list)
B. size(list)
C. lengthof(list)
D. countall(list)
ANSWER: A

What does list.pop() do by default?
A. Sorts the list
B. Removes and returns the last item
C. Deletes the variable permanently
D. Copies the list
ANSWER: B

Which method returns dictionary key-value pairs as a view?
A. items()
B. entrieslistonly()
C. pairs()
D. valueskeys()
ANSWER: A

Which statement updates the value for key 'city' in a dictionary d?
A. d('city') == 'Pune'
B. d['city'] = 'Pune'
C. d.city := 'Pune'
D. UPDATE d city 'Pune'
ANSWER: B

Which operator joins two lists into a new list?
A. +
B. .
C. &
D. ::
ANSWER: A

Which expression checks whether 5 is present in list nums?
A. nums has 5
B. 5 in nums
C. nums == 5
D. contains(nums, 5)
ANSWER: B

Which list method removes the first matching value?
A. erasefirst()
B. discard()
C. deletevalue()
D. remove()
ANSWER: D

Which statement deletes the item at index 2 from list items?
A. items.removeindex(2)
B. items.pop(2) only syntax
C. del items[2]
D. delete items 2
ANSWER: C

Which loop is commonly used to visit each item in a list?
A. loop(items)
B. for item in items:
C. repeat item of items
D. foreach items do
ANSWER: B

Which dictionary method removes a key and returns its value?
A. append()
B. extend()
C. sort()
D. pop()
ANSWER: D

What does list.extend(other) do?
A. Adds the iterable as a single nested item always
B. Adds elements from an iterable to the list
C. Deletes the original list
D. Sorts both lists
ANSWER: B

Which expression creates a list containing the elements of a and b?
A. a - b
B. a * b only
C. a + b
D. a / b
ANSWER: C

Which method clears all items from a dictionary d?
A. clear(d, keys)
B. d.clear()
C. d.removeall()
D. d.empty()
ANSWER: B

What does a dictionary key need to support?
A. Being a list always
B. Being a file path
C. Being mutable in all cases
D. Hashability
ANSWER: D

Which statement correctly checks if a key exists in dictionary data?
A. data haskey 'id'
B. 'id' inside values(data)
C. data['id'] exists
D. 'id' in data
ANSWER: D

Which syntax creates a tuple with two values?
A. [1, 2]
B. (1, 2)
C. set(1, 2)
D. {1: 2}
ANSWER: B

What is a key property of a tuple?
A. It is always sorted
B. It only stores integers
C. It cannot be indexed
D. It is immutable
ANSWER: D

Which is the correct way to create a one-item tuple?
A. {5}
B. (5)
C. (5,)
D. [5,]
ANSWER: C

Which assignment unpacks two values into two variables?
A. x = y = 1, 2 only
B. unpack x y from 1 2
C. x, y = 1, 2
D. x:y = (1,2)
ANSWER: C

What does 'a,b = b,a' do in Python?
A. Creates a dictionary
B. Swaps the values of a and b
C. Raises an error in all cases
D. Adds a and b
ANSWER: B

Which string method splits a string into a list of parts?
A. join()
B. partitionlist()
C. split()
D. separate()
ANSWER: C

Which string method joins iterable strings using a separator?
A. mergechars()
B. split()
C. join()
D. append()
ANSWER: C

Which built-in type stores unique elements?
A. str
B. list
C. set
D. tuple
ANSWER: C

Which operator combines two sets by union?
A. +
B. &
C. |
D. /

ANSWER: C

How can you create an empty set?
A. {}
B. []
C. set()
D. ()
ANSWER: C

What is a default argument in a Python function?
A. A return type
B. A required import
C. A global variable only
D. An argument value used when no value is supplied
ANSWER: D

Which call uses a named argument?
A. greet(name='Asha')
B. greet=>name
C. greet('Asha')
D. greet Asha
ANSWER: A

What does the built-in type() function report?
A. An object's type
B. The current time
C. The object's memory address only
D. A file's permissions
ANSWER: A

What does str(25) return?
A. The integer 25
B. The Boolean True
C. A list containing 25
D. The string '25'
ANSWER: D

What is dir(object) commonly used for?
A. Opening a socket
B. Changing the object's type
C. Deleting the object
D. Listing attributes available on an object
ANSWER: D

In Python, variables generally refer to what?
A. Objects
B. CPU registers directly only
C. Source-code line numbers
D. SQL tables automatically
ANSWER: A

What is a socket commonly used for?
A. Creating a tuple
B. Changing indentation
C. Sorting a list
D. Communication between processes over a network
ANSWER: D

Which Python module provides socket networking interfaces?
A. websocket_only
B. tcpip
C. socket
D. networking
ANSWER: C

Which method commonly starts a server socket listening for connections?
A. acceptall()
B. listenserveronly()
C. starttcp()

D. listen()
ANSWER: D

Which socket method waits for an incoming connection on a listening server socket?
A. accept()
B. bindonly()
C. connect()
D. sendall()
ANSWER: A

Which Python module supports regular expressions?
A. pattern
B. re
C. regexpy
D. search
ANSWER: B

Which regex symbol matches any single character except a newline by default?
A. *
B. .
C. $
D. ^
ANSWER: B

Which regex token commonly matches a word character?
A. \w
B. \n
C. \s
D. \D
ANSWER: A

Which regex token matches a whitespace character?
A. \W
B. \S
C. \s
D. \d
ANSWER: C

Which regex token matches a non-whitespace character?
A. \b
B. \w
C. \S
D. \s
ANSWER: C

Which function checks for a match at the beginning of a string?
A. re.beginning()
B. re.end()
C. re.match()
D. re.startonly()
ANSWER: C

Which function searches for a match anywhere in a string?
A. re.locate()
B. re.any()
C. re.matchallonly()
D. re.search()
ANSWER: D

What is a Python script?
A. A file containing Python code that can be executed
B. A regular expression only
C. A hardware device
D. A database table
ANSWER: A

What is a lambda expression commonly used to create?
A. A class constructor only
B. A small anonymous function

C. A database schema
D. A socket server automatically
ANSWER: B

Which functional programming tool applies a function to items in an iterable?
A. map()
B. socket()
C. input()
D. mkdir()
ANSWER: A

Which keyword defines a class in Python?
A. define
B. class
C. object
D. struct
ANSWER: B

What is an object in object-oriented programming?
A. An instance of a class
B. A SQL query
C. A directory permission
D. A Python comment
ANSWER: A

What does self conventionally refer to in an instance method?
A. A global socket
B. The parent directory
C. The current instance
D. The Python interpreter
ANSWER: C

Which method is commonly used to initialize a new instance's attributes?
A. __newclass__()
B. __start__()
C. init_object()
D. __init__()
ANSWER: D

What is encapsulation in OOP?
A. Bundling data and behavior while controlling access
B. Deleting all objects
C. Converting all data to strings
D. Sorting variables alphabetically
ANSWER: A

Which module provides tools such as getcwd() and listdir() for filesystem paths and directories?
A. os
B. statistics
C. re
D. math
ANSWER: A

Which function returns the current working directory?
A. os.pwdpath()
B. path.currentdir()
C. os.getcwd()
D. os.current()
ANSWER: C

Which function checks whether a path exists?
A. os.path.present()
B. file.isthere()
C. path.available()
D. os.path.exists()
ANSWER: D

Which Python module can inspect and change file permission bits on Unix-like systems?
A. json

B. os
C. random
D. calendar
ANSWER: B

What is an absolute path?
A. A file's contents
B. A path specified from the filesystem root or full location
C. A directory listing
D. A path relative to the current directory only
ANSWER: B

Which built-in function opens a file in Python?
A. open()
B. access()
C. readfile()
D. fileopen()
ANSWER: A

Which file mode opens a file for reading text?
A. 'w'
B. 'r'
C. 'a'
D. 'x'
ANSWER: B

Which file mode writes to a file and truncates existing content?
A. 'rb'
B. 'w'
C. 'a'
D. 'r'
ANSWER: B

Which file mode appends data to the end of a file?
A. 'r'
B. 'a'
C. 'w'
D. 'x'
ANSWER: B

Why is 'with open(...) as f:' recommended?
A. It manages closing the file automatically
B. It converts the file to a database
C. It prevents all read errors
D. It encrypts the file
ANSWER: A

Which file method reads the entire remaining content as a string?
A. closecontent()
B. write()
C. seekonly()
D. read()
ANSWER: D

Which file method writes text to an open file?
A. write()
B. split()
C. read()
D. listdir()
ANSWER: A

What does a relative path depend on?
A. The Python version only
B. The current working directory or a specified base
C. The network speed
D. The file's contents
ANSWER: B

What does a directory permission commonly control?

A. The file encoding only
B. The font used in filenames
C. Who may read, write, or traverse the directory
D. The number of CPU cores
ANSWER: C

Which socket concept identifies the local address and port for a server socket?
A. split()
B. chmod()
C. bind()
D. compile()
ANSWER: C

Which file method reads one line at a time?
A. readalllines()
B. readline()
C. appendline()
D. write()
ANSWER: B

How can you count lines in a text file using Python?
A. Use os.getcwd()
B. Use len(file_path) only
C. Call socket.listen()
D. Iterate over the file's lines and increment a counter
ANSWER: D

How can you count words in a text file simply?
A. Split the file text on whitespace and count the resulting items
B. Call file.close() twice
C. Count only newline characters
D. Use os.listdir()
ANSWER: A

Which method returns all lines of a file as a list?
A. getlinesonly()
B. readlines()
C. writeLines()
D. linecount()
ANSWER: B

What does os.listdir(path) return?
A. Network sockets
B. File contents as text
C. The absolute path of every file only
D. Names of entries in a directory
ANSWER: D

Which Python library is commonly used to make HTTP requests to retrieve web page content?
A. re
B. requests
C. datetime
D. hashlib
ANSWER: B

What should a program commonly check after an HTTP request?
A. The database schema name
B. The CPU fan speed
C. The response status or whether the request succeeded
D. The tuple length only
ANSWER: C

Which permission symbol is commonly associated with read access in Unix-like systems?
A. w
B. x
C. p
D. r
ANSWER: D

Which socket method sends bytes to a connected peer?
A. listen()
B. bind()
C. send() or sendall()
D. accept()
ANSWER: C

Why should file handles be closed when finished?
A. To create a new socket
B. To erase the file automatically
C. To change the current directory
D. To release resources and flush pending writes
ANSWER: D

Which Python module provides cryptographic hash functions such as SHA-256?
A. calendar
B. socketserveronly
C. datetime
D. hashlib
ANSWER: D

What is a cryptographic hash commonly used for?
A. Opening directories
B. Recovering the original plaintext reliably
C. Assigning IP addresses
D. Producing a fixed-length digest of input data
ANSWER: D

Which Python module provides date and time classes?
A. hashlib
B. datetime
C. os.path
D. re
ANSWER: B

Shodan is commonly associated with searching for what?
A. SQL table schemas
B. Local-only text files
C. Python list indexes
D. Internet-connected devices and exposed services
ANSWER: D

In a client-server architecture, what does a client typically do?
A. Compiles the server's operating system
B. Requests a service or resource from a server
C. Always stores the only database copy
D. Assigns every port automatically
ANSWER: B

What does a web server commonly do?
A. Receives HTTP requests and returns responses
B. Only stores tuples
C. Converts all SQL to CSV automatically
D. Only edits Python indentation
ANSWER: A

What is debugging?
A. Sorting a directory listing
B. Finding and correcting defects in a program
C. Creating a database user
D. Encrypting every network packet
ANSWER: B

Which is a useful first step when a Python program raises an exception?
A. Ignore the message and continue
B. Delete the Python installation immediately
C. Read the traceback and identify the error location
D. Rename every variable
ANSWER: C

What is exploit development in the syllabus context?
A. Designing database table names only
B. Studying and constructing code that demonstrates or uses a vulnerability
C. Formatting a disk
D. Writing comments in Python
ANSWER: B

When testing socket-based exploit code, what is an important practice?
A. Disable all monitoring
B. Ignore error handling
C. Use an authorized, isolated lab environment
D. Run it against any public host
ANSWER: C